Point Set Topology

$$
\begin{tikzcd}
A \arrow[rrr, "i_0"] \arrow[dr, "g"] \arrow[ddd, "i"'] & & & A \times I \arrow[dl, "g\times\mathrm{id}"'] \\
& B \arrow[rr] \arrow[dd, "\text{pushout}"'] & & B \times I \arrow[dl, "h"'] \arrow[dd] \\
& & Y & \\
& B \cup_g X \arrow[uur, "f"] \arrow[rr] \arrow[uu, dashed] & & (B \cup_g X) \times I \arrow[ul, dashed, "\widetilde{h}"'] \\
X \arrow[ur] \arrow[rrr, "i_0"'] & & & X \times I \arrow[ul] \arrow[uu, "i\times\mathrm{id}"'] \arrow[uull, dashed, "\bar{h}"]
\end{tikzcd}
$$

Notes for NJUPT Topology (Spring 2024)

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Based on book of M. A. Armstrong and Munkres

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Lecture 1

Topological Spaces

> *I believe that we lack another analysis properly geometric or linear which expresses location directly as algebra expresses magnitude.*
>
> —— G. W. Leibniz

1.0 Introduction to Topology

Motivation. [1] The subject of topology deals with the expressions of continuity and boundary, and studying the geometric properties of (originally: metric) spaces and relations of subspaces, which do not change under continuous deformations, regardless to other (such as in their metric properties). According to Bourbaki[2] say:

- structures defined on sets,
- maps that preserve structures.

Topology as a structure enables one to model continuity and convergence locally. More recently, in metric spaces, topologists and geometric group theorists started looking at asymptotic properties at large, which are in some sense dual to the standard topological structure and are usually referred to as coarse topology.

1.1 Basic Concepts

Before we defined the open set, we use neighbourhood to described the topological spaces.

Definition 1.1.1. A set X and $\forall\, x \in X$, there's a nonempty collection of subset of X, we called neighbourhood. If neighbourhood follows the below axioms:

- x lies in each of its neighbourhood
- The intersection of two neighbourhoods of x is itself i.e. a neighbourhood of x.
- If N is a neighbourhood of x and if $U \subset X$ contains N, then U is neighbourhood of X.
- If N is a neighbourhood of x and if $\mathring{N} := \{z \in N : N \text{ is a neighbourhood of } Z\}$, then $\mathring{N}$ is a neighbourhood of x.

Then we called X is a **Topological Space**.

Remark. The $\mathring{N}$ is called interior of N.

Remark. A collection of neighbourhood such that the all of the axioms to every points in X is called **Topology** on X

Definition 1.1.2 (Continuous)**.** Let X and Y are topological spaces, a map $f : X \to Y$ is continuous if $\forall\, x \in X$ and for any neighbourhood N of image $f(X)$ in Y such that $f^{-1}(N)$ is neighbourhood of x in X.

[1] Excerpts from nLab —https://ncatlab.org/nlab/show/topology
[2] See in Éléments de mathématique — Topologie Générale

Example (Subspace topology)**.** Let X be a topological space and let Y be a subset of Y i.e. $Y \subset X$, define a topology τ_y on the set Y. Let $y \in Y$, take collection of its neighbourhoods in topological space X and intersect each of these neighbourhood with Y. Obviously, it satisfy the four axioms, so Y is a Topological Space. ◇

Proof. Let N be neighbourhood, take $N \cap Y$, it's neighbourhood of Y, let $y \in Y$, hence $N \cap Y$ also be a neighbourhood of y, $N \subset X$, so $y \in N \subset X$ implies $y \in N \cap Y$, that's satisfy axiom (a). For (b), suppose N and M are neighbourhood in X, let $N \cap Y$ and $M \cap Y$ are neighbourhood of $y \in Y$, take

$$(N \cap M) \cap Y = (N \cap Y) \cap (M \cap Y)$$

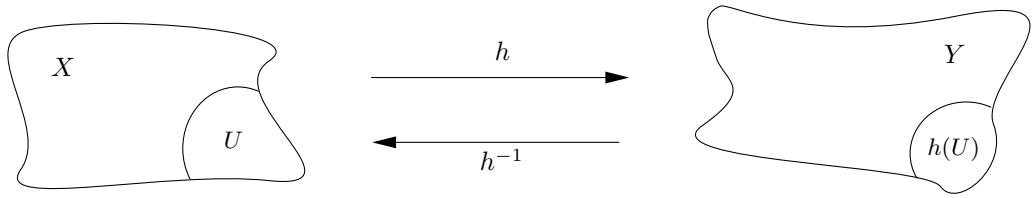
That's a neighbourhood of $y \in Y$. So satisfy axiom (b). For (c). Let U be subset of Y, $N \cap Y$ is neighbourhood of $y \in Y$, let $N \cap Y \subseteq U$, so $N \cup U$ is subset of X contains N, so $N \cup U$ is neighbourhood in X implies

$$(N \cup U)Y = (N \cap Y) \cup (U \cap Y) = U \cap Y = U$$

Thus U is neighbourhood of $y \in Y$ satisfy axiom (c). For (d), let $N \cap Y$ is neighbourhood of $y \in Y$, let $(N \cap Y)^\circ$ is interior of $N \cap Y$. If $a \in (N \cap Y)^\circ \iff N \cap Y$ is neighbourhood of a, hence, N is neighbourhood of $a \in Y \subset X$, so $a \in \mathring{N} \subset X$. Therefore, $(N \cap Y)^\circ = \mathring{N} \cap Y$. So $(N \cap Y)^\circ$ is neighbourhood of $y \in Y$ satisfy axiom (d). Hence, it's a topology. ■

Remark. We called the τ_y is subspace topology and Y is subspace.

Definition 1.1.3 (Homeomorphism)**.** Let X and Y are topological spaces, define map $h : X \to Y$, if h is bijection and continuous and h^{-1} also continuous, then we called h is homeomorphism. We usually denoted as $X \cong Y$.



Remark. In view of category theory, Topological spaces and all continuous map become a category, the homeomorphism can seen as a isomorphism. Some time we called homeomorphism be a Topological isomorphism.

Notation. Many authors strictly distinguish the symbols $\cong$ and $\simeq$ use the former to denote homeomorphisms and the latter to refer to homotopy equivalences. In this lecture, We use either symbol (but mostly $\cong$).

Properties. Let X, Y be topological spaces. If $f : X \to Y$ is a homeomorphism with inverse continuous function g, then

- Also g is a homeomorphism, with the inverse continuous function f.
- The underlying function of sets $f : X \to Y$ of a homeomorphism f is necessarily a bijection, with inverse bijection g.

Caveat 1. But beware that not every continuous function which is bijective on underlying sets is a homeomorphism. While an inverse function g will exists on the level of functions of sets, this inverse may fail to be continuous: See the below example.

Example (Open interval homeomorphic to the real line)**.** The open interval $(-1, 1)$ is homeomorphic to real line $\mathbb{R}$, i.e. define map

$$\begin{aligned} f : (-1, 1) &\longrightarrow \mathbb{R} \\ x &\longmapsto \frac{x}{\sqrt{1 - x^2}} \end{aligned}$$

The inverse of f is clearly continuous

$$\begin{aligned} f^{-1} : \mathbb{R} &\longrightarrow (-1, 1) \\ x &\longmapsto \frac{x}{\sqrt{1 + x^2}} \end{aligned}$$

Furthermore, every open ball is homeomorphic to $\mathbb{R}^n$ ◇

Example. Let $\mathbb{C}$ be complex plane and have interval in real line $[0,1)$, give the sub-topology to this interval and define continuous map

$$\begin{aligned} f : [0,1) &\longrightarrow \mathbb{C} \\ x &\longmapsto e^{2\pi i x} \end{aligned}$$

f is clearly bijiective, but f^{-1} is not continuous. So it's not homeomorphism. $\diamond$

Example (Stereographic Projection)**.** Let $\mathbb{S}^2$ is unit sphere in $\mathbb{R}^3$, take $\mathbb{S}^2 \setminus \{N\}$, where N is north pole, we define a map

$$P : \mathbb{S}^2 \setminus \{N\} \longrightarrow \mathbb{R}^2$$

Let $\mathbb{S}^2 = x^2 + y^2 + z^2 = 1$ and $N = (0,0,1)$, let $x = (\xi, \eta, \zeta)$, we have formula

$$Nx = \begin{cases} \xi = tx \\ \eta = ty \\ \zeta = 1 - t \end{cases}$$

We substitute Nx to $\mathbb{S}^2 = x^2 + y^2 + z^2 = 1$, then we have

$$\begin{aligned} t^2x^2 + t^2y^2 + t^2 - 2t &= 0 \\ \Longrightarrow t^2(x^2 + y^2 + 1) &= 2t \\ \Longrightarrow t(x^2 + y^2 + 1) &= 2 \end{aligned}$$

Let $\|P(x)\| = \sqrt{x^2 + y^2}$, then the above equation can rewrite

$$t(\|P(x)\|^2 + 1) = 2 \Longrightarrow t = \frac{2}{1 + \|P(x)\|^2}$$

Let $z = \|P(x)\|^2$, substitute to Nx, we can get

$$\begin{cases} \xi = \frac{2x}{1+z} \\ \eta = \frac{2y}{1+z} \\ \zeta = \frac{z-1}{z+1} \end{cases}$$

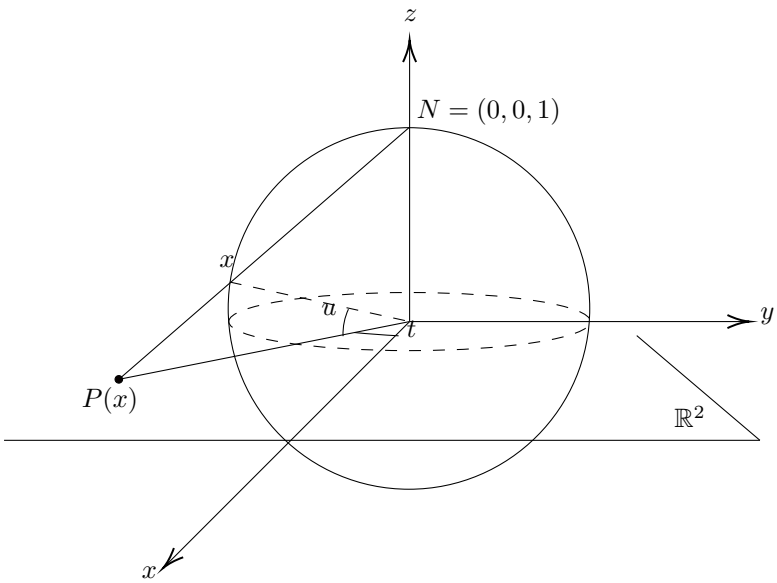
Let $1 - \zeta = \frac{2}{1+z}$, we substitute above, we have:

$$x = \frac{\xi}{1-\zeta}, \quad y = \frac{\eta}{1-\zeta}$$

Take inverse of P

$$P^{-1} : \mathbb{R}^2 \longrightarrow \mathbb{S}^2 \setminus \{N\}$$

Clearly, P is one to one and it's a homeomorphism.



◇

Remark. The function which sends a point $x \in \mathbb{S}^n \backslash \{N\} \subset \mathbb{R}^{n+1}$ to the intersection of the line through p and x with the equatorial hyperplane is a homeomorphism which is given in terms of ambient coordinates by

$$\begin{aligned} \mathbb{R}^{n+1} \supset \mathbb{S}^n \backslash (1, 0, \cdots, 0) &\longrightarrow \mathbb{R}^n \subset \mathbb{R}^{n+1} \\ (x_1, x_2, \cdots, x_{n+1}) &\longmapsto \frac{1}{1 - x_1}(0, x_2, \cdots, x_{n+1}) \end{aligned}$$

Remark. The Stereographic Projection also a classical example of a topological manifold.

Example. The radial projection

$$\begin{aligned} \pi : \mathbb{S}^1 \setminus \{0\} &\longrightarrow \mathbb{S}^2 \\ x &\longmapsto \frac{x}{\|x\|} \end{aligned}$$

is a homeomorphism. ◇

1.2 Open sets and Closed sets

In this section, we'll define open set and used language of open set to rewrite the definition of topological space. Let X be a topological space and subset $O \subseteq X$ is open if it's a neighbourhood of each points. By axiom(c), the union of open set is open and by axiom (d), finite intersection of open set is open and whole space and $\emptyset$ is open. Here's the definition below:

Definition 1.2.1 (Topological space)**.** A topological space is (X, τ) where X is a set and τ a family of subsets of X, called open sets, such that:

- $\emptyset, X \in \tau$
- Any union of open set is open i.e. $\bigcup_{i \in I} U_i \in \tau$ whenever $U_i \in \tau$ for all i
- Any finite intersection of open set is open i.e. $\bigcap_{i<n} U_i \in \tau$ whenever $U_i \in \tau$ for all i

A set with topology called topological space, denoted as (X, τ).

Remark. The Definition 1.1.1 is equivalence to Definition 1.2.1, i.e. (1) - (3) satisfy topological axioms (a) - (d).

Proof. Given a point $x \in X, N$ is subset of X and N is neighbourhood if $\exists$ open set O such that $x \in O \subseteq N$. Every points at least have one neighbourhood $\Longrightarrow$ the whole X, clearly satisfy axiom (a) and axiom (c). If N_1, N_2 are neighbourhood of x, we can find open sets O_1 and O_2 such that $x \in O_1 \subseteq N_1$ and $x \in O_2 \subseteq N_2$. Let $x \in O_1 \cap O_2$ implies $X \in O_1 \cap O_2 \subseteq N_1 \cap N_2$, by the $O_1 \cap O_2$ is open. Hence, $N_1 \cap N_2$ is neighbourhood of x such that axiom (b). Suppose N is a neighbourhood of x, let $\mathring{N}$ be interior of N. Let open set O such that $x \in O \subseteq N$, now O is open and be neighbourhood of each points, so O contained $\mathring{N}$. Therefore, $\mathring{N}$ is neighbourhood of x satisfy axiom (d). ■

Definition 1.2.2. The open sets of topology on $\mathbb{R}^n$ foloows: A set U is open if $\forall\ x \in U$, $\exists\ \varepsilon > 0$ such that $\mathbb{B}_\varepsilon(x) \subset U$.

Example (Euclidean Topology)**.** Let $\mathbb{R}$ is set of real number, define $\tau_e := \{U : U \text{ is union of open intervals}\}$ This oopen interval can infinite, finite or 0. Hence, $\phi \in \tau_e$, τ_e is topology on $\mathbb{R}$ called Euclidean Topology, denoted as $\mathbb{E}^1 = (\mathbb{R}, \tau_e)$ ◇

Example (Trivial Topology)**.** Let X be a set, only $\{X, \emptyset\}$ be topology on X called Trivial Topology. ◇

Example (Discrete Topology)**.** Let X be set, every subset U of X is open set, define $\tau_d = \{U : U \subset X\}$ is called Discrete Topology. ◇

Example. Let X be infinite set, $\tau_f := \{A^c : A \text{ is finite subset of } X\} \cup \{\emptyset\}$, τ_f is topology on X called **cofinite topology** or **finite complement topology**. ◇

Example. Let X be infinite uncountable set, $\tau_c := \{A^c : A \subseteq X, A \text{ is countable}\} \cup \{\emptyset\}$, τ_c is topology on X, called **cocountable topology** or **countable complement topology** ◇

Definition 1.2.3 (Subspace)**.** Let X be topological space and $Y \subset X$, the open sets of subspace or induced topology on Y are obtained simply by intersecting all of the open sets of X with Y, i.e.

$$\tau := \{U : U = O \cap Y, O \text{ is open in} X\}$$

τ is subspace topology.

Example (n-dim sphere)**.** The n-dimensional sphere is subspace of $\mathbb{R}^n$, denoted as

$$\mathbb{S}^n := \left\{x \in \mathbb{R}^{n+1} : x_1^2 + \cdots + x_{n+1}^2 = 1\right\}$$

The n-dimensional closed disk D^n is defined by subspace of $\mathbb{R}^n$, denoted as

$$D^n := \left\{x \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 \leq 1\right\}$$

The n-dimensional open disk is

$$\left\{x \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 < 1\right\}$$

◇

Example. The open subsets of a closed interval $[0, 1] \subset \mathbb{R}$ regarded as a topological of the real line equipped with its Euclidean metric topology. ◇

Definition 1.2.4. Let $\{(X_i, \tau)\}_{i \in \mathcal{I}}$ be a family of topological spaces. We say their sum of disjoint union topology is:

- As a set, it's $\bigsqcup_{i \in \mathcal{I}} X_i$ (disjoint union of set).
- A subset of $\bigsqcup_{i \in \mathcal{I}} X_i$ is open if and only if the its intersection with every X_i is open.

Convention. We use notation $\bigsqcup_{i \in \mathcal{I}} X_i$ to represent the topological sum or disjoint union topology.

Example. The disjoint union of point set $\{*\}$ is discrete topological space. The subspace topology of set $(\{0\} \cup \{1, 1/2, 1/3, \ldots\})$ on $\mathbb{R}$ is not disjoint union topology of set $(\{0\}, \{1\}, \{1/2\}, \ldots)$ ◇

Proposition 1.2.1 (Universal property of disjoint union spaces)**.** Let X_1, X_2 be topological spaces. Consider their disjoint union space $X_1 \sqcup X_2$. By Definition 1.2.4, the two inclusions into the disjoint union space are clearly continuous functions

$$X_1 \xrightarrow{i_1} X_1 \sqcup X_2 \xleftarrow{i_2} X_2$$

Let Y be any other topological space. Define the two continuous functions $f_1 : X_1 \to Y$ and $f_2 : X_2 \to Y$. Then exist unique map $f : X_1 \sqcup X_2 \to Y$ such that $f_1 = f \circ i_1$ and $f_2 = f \circ i_2$. we have commutative diagram:

$$\begin{tikzcd}
X_1 \arrow[r, hook, "i_1"] \arrow[dr, "f \circ i_1"'] & X_1 \sqcup X_2 \arrow[d, dashed, "\exists! f"] & X_2 \arrow[l, hook', "i_2"'] \arrow[dl, "f \circ i_2"] \\
& Y &
\end{tikzcd}$$

Proof. Given f_1 and f_2, define $f : X_1 \sqcup X_2 \longrightarrow Y$ given by

$$f(\omega) := \begin{cases} f_{X_1}(\omega) & \text{if } \omega \in X_1 \\ f_{X_2}(\omega) & \text{if } \omega \in X_2 \end{cases}$$

Clearly satisfy $f_1 = f \circ i_1$ and $f_2 = f \circ i_2$. Now, we need to prove it's unique. Let any element $\omega \in X_1 \sqcup X_2$ is in one of the summands:

$$\omega = \begin{cases} i_1(\omega) & \text{if } \omega \in X_1 \\ i_2(\omega) & \text{if } \omega \in X_2 \end{cases}$$

Therefore, take function $g : X_1 \sqcup X_2 \longrightarrow Y$ can be written as

$$g(\omega) = \begin{cases} g(i_1(\omega)) = (g \circ i_1)(\omega) & \text{if } \omega \in X_1 \\ g(i_2(\omega)) = (g \circ i_2)(\omega) & \text{if } \omega \in X_2 \end{cases}$$

So, g is determined by $g \circ i_1$ and $g \circ i_2$. Hence $g = f$, then f is unique. ■

Remark. The disjoint union also called **coproduct**.

Corollary 1.2.1. Let $(X_i)_{i \in \mathcal{I}}$ be a family of topological spaces. The disjoint union space $X := \bigsqcup_{i \in \mathcal{I}} X_i$, together with the canonical injections, can be characterized by the following universal property: If Y is a topological space, and $f_i : X_i \to Y$ is a continuous map for each $i \in l$, then there exists precisely one continuous map $f : X \to Y$ such that the following set of commutative diagrams:

$$\begin{tikzcd} X \arrow[dr, "f"] & \\ X_i \arrow[u, hook, "\phi_i"] \arrow[r, "f_i"] & Y \end{tikzcd}$$

Proof. Let $(X_i)_{i \in I}$ be a family of topological spaces and $X := \sqcup_{i \in I} X_I = \cup_{i \in I} (X_i \times \{i\})$ be their disjoint union. For each $i \in I$ we have the injection $\phi_i : X_i \to X$ given by $\phi_i(x) = (x, i)$. The topology on X is given by

$$\left\{ U \subseteq X \mid \phi_i^{-1}(U) \text{ is open in } X_i \text{ for every } i \in I \right\}.$$

Clearly this makes each ϕ_i a continuous function. Let Y be a topological space and $(f_i)_{i \in I}$, with $f_i : X_i \to Y$ for each $i \in I$, be a family of continuous functions. We wish to show that there is a unique continuous function $f : X \to Y$ such that $f_i = f \circ \phi_i$ for every $i \in I$.

In order for a function $f : X \to Y$ to satisfy $f_i = f \circ \phi_i$, we must have $f(x, i) = f(\phi_i(x)) = f_i(x)$ for every $(x, i) \in X$. There is only one function that does this, so it remains to show that the function $f : X \to Y$ defined by $f(x, i) = f_i(x)$ is continuous. Suppose U is an open subset of Y. To conclude that $f^{-1}(U)$ is open in X, we must show that $\phi_i^{-1}\left(f^{-1}(U)\right)$ is open in X_i for each $i \in I$. Fixing $i \in I$, we have

$$\phi_i^{-1}\left(f^{-1}(U)\right) = (f \circ \phi_i)^{-1}(U) = f_i^{-1}(U),$$

which is open because f_i was assumed to be continuous. This completes the proof. ■

Theorem 1.2.1. The canonical inclusions $X_1 \hookrightarrow X_1 \sqcup X_2 \hookleftarrow X_2$ are both open and closed.

Definition 1.2.5. Suppose that τ and τ' are two topologies on a given set X. If $\tau' \subset \tau$, we say that τ' is finer than τ, if $\tau' \supset \tau$, if τ' properly contains τ; we say τ' is strictly finer than τ. We also say τ is coarser than τ' or strictly coarser. In this two respective situations, We say τ is comparable with τ' if either $\tau' \supset \tau$ or $\tau \supset \tau'$.

Remark. Some times, if $\tau' \supset \tau$, we say τ' is **larger** tan τ and τ is **smaller** than τ'. Also, we could use **weak** and **strong** to represent.

Proposition 1.2.2 (Universal property of subspace)**.** Let X be a topological space and Y is a subset of X, define $i : Y \to X$ be an injective continuous function between topological spaces. Then this is a subspace inclusion precisely if it satisfies the following universal property:

- For any topological space Z, exist a function $f : Z \to Y$ is continuous, and if the composition such that the $i \circ f$ is continuous as a function to X, we have commutative diagram:

$$\begin{tikzcd} Z \arrow[r, "f"] \arrow[dr, "i\circ f"'] & Y \arrow[d, hook, "i"] \\ & X \end{tikzcd}$$

Proof. Let τ_Y be the subspace topology on Y. Let (Z, τ_Z) be any topological space and let $f : Z \to Y$. We have claim,

Claim. $f : Z \to Y$ is continuous if and only if $i \circ f : Z \to X$ is continuous.

Proof of Claim. ($\Longrightarrow$): If f is continuous, by the proposition i is continuous, clearly $i \circ f : Z \to X$ is continuous.
($\Longleftarrow$): If $i \circ f : Z \to X$ is continuous. Let U be an arbitrary open set in Y. Then $U = i^{-1}(V)$, where V is some open set in X. Since $i \circ f$ is continuous, then $(i \circ f)^{-1}(V) \subset Z$ is open in Z. Since $(i \circ f)^{-1}(V) = f^{-1}(U)$, we conclude the $f^{-1}(U)$ is open, so $f : Z \to Y$ is continuous. ■

By claim, proof complete. ■

Corollary 1.2.2. If Y is equipped with any topology having the universal property, then that topology must be the subspace topology. We have the following commutative diagram:

$$\begin{tikzcd} (Y, \tau_Y) \arrow[r, "\mathrm{id}_Y"] \arrow[dr, "i\circ \mathrm{id}_Y"'] & (Y, \tau') \arrow[d, "i"] \\ & X \end{tikzcd}$$

Proof. Now assume that τ' is a topology on Y and that τ' has the universal property. We have claim

Claim. Topology τ' equals the subspace topology τ_Y.

Proof of Claim. Suppose that when Y has topology τ', then for every topological space (Z, τ_Z) and for any function $f : Z \to Y$ is continuous if and only if $i \circ f$ is continuous. In particular, if let (Z, τ_Z) be (Y, τ_Y) where τ_Y is the subspace topology on Y. Let $\mathrm{id}_Y : Y \to Y$ be identity map, then we have the following commutative diagram:

$$\begin{tikzcd} (Y, \tau_Y) \arrow[r, "\mathrm{id}_Y"] \arrow[dr, "i\circ \mathrm{id}_Y"'] & (Y, \tau') \arrow[d, "i"] \\ & X \end{tikzcd}$$

Since we know the $i \circ \mathrm{id}_Y = i : Y \to X$ is continuous when Y has the subspace topology τ, by the universal property, we have id_Y is continuous. That implies that the subspace topology. τ_Y is finer than τ', i.e. $\tau' \subset \tau_Y$, now we need show $\tau_Y \subset \tau'$, let $f = \mathrm{id}_Y : (Y, \tau') \to (Y, \tau')$. Since id_Y is continuous, we must have $i \circ \mathrm{id}_Y = i : Y \to X$ is continuous. So we can use the commutative diagram to represent the relation:

$$\begin{tikzcd} (Y, \tau') \arrow[r, "\mathrm{id}_Y"] \arrow[dr, "i\circ \mathrm{id}_Y"'] & (Y, \tau') \arrow[d, "i"] \\ & X \end{tikzcd}$$

Since the subspace topology τ_Y is the coarsest topology on Y for which $i : Y \to X$ is continuous, we conclude that τ_Y is coarser than τ', i.e. $\tau_Y \subset \tau'$. Therefore, the conclusion is that $\tau' = \tau_Y$. ■

By claim, proof complete. ■

Definition 1.2.6 (Closed set)**.** A subset of a topological space is close if its complement is open.

Proposition 1.2.3. Topological space is closed such that

1). X and $\emptyset$ are closed

2). Any intersection of any family of closed set is closed

3). The union of any finite family closed set is closed

Proof. 1). By Definition 1.2.1, X is the whole set and the complement of X is $X^c = \emptyset$, the $\emptyset$ is open, hence, the X is closed. Similarly, $\emptyset^c = X$, X is open so $\emptyset$ is closed.
2). Any union of open set is open, let $A := \bigcup_{\lambda \in \mathbb{A}} A_\lambda$ be union of the open sets, X is whole set, according to De Morgan law, we have

$$\left(\bigcup_{\lambda \in \mathbb{A}} A_\lambda\right)^c = X \setminus \bigcup_{\lambda \in \mathbb{A}} A_\lambda = \bigcap_{\lambda \in \mathbb{A}} A_\lambda^c$$

Hence, $\bigcap_{\lambda \in \mathbb{A}} A_\lambda^c$ is closed.
3). Similarly to 2). finite intersection of open sets is open, let $A := \bigcap_{\lambda \in \mathbb{A}} A_\lambda$ be finite intersection of open sets, according to De Morgan law, we have

$$\left(\bigcap_{\lambda \in \mathbb{A}} A_\lambda\right)^c = X \setminus \bigcap_{\lambda \in \mathbb{A}} A_\lambda = \bigcup_{\lambda \in \mathbb{A}} A_\lambda^c$$

Hence, $\bigcup_{\lambda \in \mathbb{A}} A_\lambda^c$ is closed. ■

Definition 1.2.7 (Limit point)**.** Let A be a subset of topological space X and called a point p of X be a limit point of A if every neighbourhood of p contains at least one point of $A \setminus \{p\}$

Remark. Such a point may or may not in A.

Example. Take X to be the real line $\mathbb{R}$, let A consist of the points $1/n$, $n = 1, 2, \cdots$. Then A has exactly one limit point, i.e. the origin. ◇

Example. Take X to be the real line $\mathbb{R}$, take $A = [0, 1)$, then each point of A is a limit point of A and in addition 1 is a limit point of A. ◇

Example. Let X be $\mathbb{E}^3$ and let A consist of those all of whose coordinates are rational. Then every point of $\mathbb{E}^3$ is a limit point of A. ◇

Example. Let $A \subseteq \mathbb{E}^3$ be the set of points which have integer coordinates. Then A don't have limit points. ◇

Theorem 1.2.2. A set is closed if and only if contains all its limit points.

Proof. ($\Longrightarrow$): If A is closed, $A^c = X \setminus A$ is open, since an open set is a neighbourhood of each of its points, no points of $X \setminus A$ can be a limit points of A. Thus A contains all of its limit points.
($\Longleftarrow$): Suppose A contains all its limit points and let $x \in X \setminus A$. If x is not a limit point of A, so we can find a neighbourhood N of x which don't intersect with A. Hence, $N \subset X \setminus A$ and $X \setminus A$ is neighbourhood of each points. Therefore, $X \setminus A = A^c$ is open, A is closed. ■

Definition 1.2.8 (Closure)**.** Let A is a subset of topological space X, every limit points of A can be a set, we called <u>derived set</u>, usually denoted as A', we define set $\bar{A} := A \cup A'$ is <u>closure</u> of A.

Proposition 1.2.4. A set is closed if and only if equal its closure.

Proof. ($\Longrightarrow$): A is closed, then by Definition 1.2.8, we know $\bar{A} := A \cup A'$, then we have $A \subseteq \bar{A}$, we have the following claim:

Claim. A is closed if and only if $A \supset A'$.

Proof of Claim. ($\Longrightarrow$): Let A is closed, we just need prove $x \in A' \Longrightarrow x \in A$, we consider the contrapositive i.e. $x \notin A' \Longrightarrow x \notin A$. Since A is closed, then A^c is open and $x \notin A \Longrightarrow x \in A^c$, hence, there exist r such that $\mathbb{B}_r(x) \subset A^c$, so $\mathbb{B}_r(x) \cap A = \emptyset$. Thus x is not a limit point of A, i.e. $x \notin A'$, that's equivalence $x \in A' \Longrightarrow x \in A$. Therefore $A \supset A'$.
($\Longleftarrow$): If $A \supset A'$, we want to prove A^c is open, let $x \in A^c$, since $A \supset A'$ implies x is not a limit point of A, so exist r s.t. $\mathbb{B}_r(x) \cap A \setminus \{x\} = \emptyset$, hence, $x \in A^c$, then $x \in A$, then $\mathbb{B}_r(x) \cap A \neq \emptyset) \Longrightarrow \mathbb{B}_r(x) \subset A^c$. Hence, A^c is open. ■

Hence, by claim proof ($\Longrightarrow$), $x \in A' \Longrightarrow x \in A$, then $\bar{A} \subset A$, then $A = \bar{A}$.
($\Longleftarrow$): If $A = \bar{A}$, then by Definition 1.2.8, $\bar{A} := A \cup A'$, notice that $A \supset A'$, by claim, A is closed. ■

Definition 1.2.9 (Boundary(or Frontier))**.** Let X be topological space, $A \subset X$, the boundary of A is the intersection of closure of A with closure of $X \setminus A$, i.e. take those points of X which not belong to the $\mathring{A}$ or $(X \setminus A^\circ$, we can use math notation represent as

$$\partial A := \overline{A} \cap \overline{(X \setminus A)}$$

Remark. We have a equivalence definition of boundary is $\partial A := \overline{A} \setminus \mathring{A}$
Example. We have an open interval $(0,1) \subset \mathbb{R}$, then $\overline{A} = [0,1], \mathring{A} = (0,1), \partial A = \{0,1\}$ ◇

Lemma 1.2.1. Let X be topological space, A is subset of X, then $\overline{\overline{A}} = \overline{A}$.

Proposition 1.2.5. The closure $\overline{A}$ is closed.

Proof. Let x be limit point of $\overline{A}$, then exist an open set U contains x in $\overline{A}$, by definition, $\overline{A} = A \cup A'$, then we have two cases:

(i) U contain infinite points in A, then x is limit point of A, i.e. $x \in A' \subset \overline{A}$.

(ii) U contain infinite points in A', then x is limit point of A'. A' is closed. Hence, $x \in A' \subset \overline{A}$.

By (i) (ii), we have $x \in \overline{A}$. Therefore, $\overline{A}$ is closed. ■

Remark. For another way to proof the proposition, we can use Proposition 1.2.4 and Lemma 1.2.1.

Proposition 1.2.6. The boundary ∂A is closed.

Proof. We know $\partial A = \overline{A} \cap \overline{(X \setminus A)}$, by Proposition 1.2.5, $\overline{A}$ and $\overline{(X \setminus A)}$ is closed and by Proposition 1.2.3. Trivial. ■

Definition 1.2.10. Let X be topological space. A is subset of X, if $(\overline{A})^\circ = \emptyset$, we say A is nowhere dense in X. The countable union of **nowhere dense** subset we called **first category**; if $\overline{A} = X$, we say A **dense** in X if and only if any open set $O \subset X$ such that $O \cap A \neq \emptyset$.

Example. The rational number $\mathbb{Q}$ is dense subspace under standard topology of $\mathbb{R}$, but $\mathring{\mathbb{Q}} = \emptyset$. ◇

Proof. According property of $\mathbb{R}$: any open interval contain rational number, also contain irrational numbers. That's implies $\mathbb{Q}$ is dense in $\mathbb{R}$ and $\mathring{\mathbb{Q}} = \emptyset$. In fact, if $x \in \mathbb{Q}$ is interior, then exist neighbourhood $(x-\varepsilon, x+\varepsilon) \subset \mathbb{Q}$ in $\mathbb{R}$. We know, the open interval must have irrational numbers, contradiction ! ■

1.3 Topological basis

Definition 1.3.1 (Basis)**.** Let X be topological space, $\mathcal{B}$ is collection of open sets such that every open sets are a union of members of $\mathcal{B}$. Then $\mathcal{B}$ is called a **basis** for the topology and elements of $\mathcal{B}$ called **basic open set**. i.e. given point $x \in X$, exist neighbourhood N of x, there exist an element $\mathscr{B}$ of $\mathcal{B}$ such that $x \in \mathscr{B} \subseteq \mathcal{B}$.

Example. For $\mathbb{R}$, a set of all open intervals is a basis. ◇

Example. For $\mathbb{R}$, a set of all open intervals which have rational endpoints is a smaller basis, it's countable. ◇

Proposition 1.3.1. A collection B of subsets of X is a base for some topology on X if an only if these conditions are met:

- The elements of B cover X;
- For any $U, V \in B$ and any point $x \in U \cap V$ there is a $W \in B$ such that $W \subseteq U \cap V$ and $x \in W$.

Theorem 1.3.1. Let β be a nonempty collection of subsets of X, if the intersection of any finite number of members of β is always in β, and if $\bigcup_{i \in I} \beta_i = X$, then β is a basis for a topology on X.

Proof. We first show that the finite intersection of β_i is in topology τ. For $n = 1$, trivial, we suppose for $n-1$ it's true; then for n, we have

$$\beta_1 \cap \cdots \cap \beta_n = (\beta_1 \cap \cdots \cap \beta_{n-1}) \cap \beta_n$$

By hypothesis, $\beta_1 \cap \cdots \cap \beta_{n-1}$ is belong to tau, then $\beta_1 \cap \cdots \cap \beta_n$ is belong to τ. Next, we take $\beta_1, \beta_2 \in \beta$, given $x \in \beta_1 \cap \beta_2$, hence, we have a neighbourhood $\mathcal{B}_1$ of x such that $\mathcal{B}_1 \subset \beta_1$, similarly, we have neighbourhood $\mathcal{B}_2$ of x such that $\mathcal{B}_2 \subset \beta_2$, can deduce the rest from this, then by the $\bigcup_{i \in I} \beta_i = X$, we got a $\mathcal{B} = \bigcup_{i \in I} \mathcal{B}_i$ cover X. By Proposition 1.3.1, β is basis. ■

Lemma 1.3.1. Let X be a set, let $\mathcal{B}$ be a basis of topology on X. Then τ equals the collection of all unions of elements of $\mathcal{B}$.

Proof. Given a collection of elements of $\mathcal{B}$, they are also elements of τ. Because τ is a topology, their union is in τ. Conversely, given $U \in \tau$, choose for each $x \in U$ an element $\mathscr{B}_x$ of $\mathcal{B}$ such that $x \in \mathscr{B}_x \subset U$. Then $U = \bigcup_{x \in U} \mathscr{B}_x$. Proof complete. ■

Lemma 1.3.2. Let (X, τ) be a topological space. Suppose that $\mathcal{C}$ is a collection of open sets of X such that for each open set U of X and each x in U, there is an element $C \in \mathcal{C}$ such that $x \in C \subset U$. Then $\mathcal{C}$ is the basis for the topology of X.

Example (Standard Topology)**.** If $\mathcal{B}$ is the collection of all open intervals in the real line

$$(a, b) = \{x : a < x < b\}$$

The topology generated by $\mathcal{B}$ is called the standard topology on $\mathbb{R}$. ◇

Lemma 1.3.3. Let $\mathcal{B}$ and $\mathcal{B}'$ be basis for topologies τ and τ', respectively, on X. Then, the following are equivalent:

1. τ' is finer than τ
2. For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x, there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Definition 1.3.2 (Subbasis)**.** A subbasis for X is a subcollection $\mathcal{S} \subset \tau$ whose members are called subbasic open subsets such that every open subset is a union of finitary intersections of subbasic ones.

Claim. The collection of $\mathcal{B}$ of all finite intersections of elements of $\mathcal{S}$ is a basis.

Proof of Claim. By Lemma 1.3.1, given $x \in X$, it belongs to an element of $\mathcal{S}$ and hence to an element of $\mathcal{B}$. Let

$$\mathscr{B}_1 = \bigcap_{i=1}^{m} S_i \quad \text{and} \quad \mathscr{B}_2 = \bigcap_{j=1}^{n} S'_j$$

be two elements of $\mathcal{B}$. The intersection is :

$$\mathscr{B}_1 \cap \mathscr{B}_2 = \left(\bigcap_{i=1}^{m} S_i\right) \cap \left(\bigcap_{j=1}^{n} S'_j\right).$$

Clearly, it's also a finite intersection of elements of $\mathcal{S}$, so it's in $\mathcal{B}$. ■

Proposition 1.3.2. Every collection $\mathcal{S}$ of subsets of X is a subbasis for some topology on X.

Proposition 1.3.3. Let $f : X \to Y$ is a map between topological spaces. $\mathcal{S}$ is subbasis of Y. Then f is continuous if an only if $\forall\, U \in \mathcal{S}$, $f^{-1}(U)$ is open set.

Proof. ($\Longrightarrow$): If f is continuous at each $x \in X$, for every $x \in X$, we have $f(x) = y \in Y$. since $\mathcal{S}$ is subbasis of Y, then $Y = \bigcup_{U \in \mathcal{S}} U$, so $y = f(x) \in \bigcup_{U \in \mathcal{S}} U$. Hence $x \in f^{-1}\left(\bigcup_{U \in \mathcal{S}} U\right)$ is open in X.
($\Longleftarrow$): Let $W \subseteq Y$ be any open set, if $\forall\, U \in \mathcal{S}$, $f^{-1}(U)$ is open set. Define a basis of the topology on Y be

$$\mathcal{S}' := \left\{\bigcap_{i=1}^{n} U_i : i \in \mathbb{N}, U_1, \cdots, U_n \in \mathcal{S}\right\}$$

Since intersections commute with preimages, $f^{-1}(U)$ is open for $U \in \mathcal{S}'$, and preimages commute with unions, so that f is continuous. ■

Lecture 2

Continuity of Topological Spaces

2.1 Continuous Function

Definition 2.1.1 (Open/Closed map)**.** Let X and Y be topological spaces, define a continuous function $f : X \to Y$ is called:

- We say f is open map if the image under f of an open subset of X is an open subset of Y.
- We say f is closed map if the image under f of a closed subset of X is a closed subset of Y.

Claim. Let X, Y be a topological spaces, function $f : X \to Y$ is continuous if and only if the inverse image of each set of Y

Proof of Claim. ($\Longrightarrow$): If $f : X \to Y$ is continuous if for each point $x \in X$ and each neighbourhood N of $f(X) \subset Y$, the $f^{-1}(N)$ is neighbourhood of $x \in X$, since f is continuous, then $\exists$ open set $O \subset Y$, so O is neighbourhood of each points, by Definition 2.1.1, $f^{-1}(O)$ is open in X.
($\Longleftarrow$): For all $x \in X$, $f(X) \subset Y$, let V is neighbourhood of $f(X)$, Let $U = f^{-1}(V)$, hence, U is neighbourhood of x, by Definition 2.1.1, f is continuous. ■

Theorem 2.1.1. The composition of two continuous map is continuous.

$$\begin{tikzcd} X \arrow[r, "f"'] \arrow[rr, bend left, "g \circ f"] & Y \arrow[r, "g"'] & Z \end{tikzcd}$$

Proof. Let X, Y, Z be topological spaces, define $f : X \to Y$ and $g : Y \to Z$ and both continuous, let open neighbourhood O of Z, take $(g \circ f)^{-1}(O) = f^{-1}(g^{-1}(O)$, because g is continuous, then by claim g^{-1} is open in Y. Similarly, since f is continuous, then $f^{-1}(g^{-1}(O))$ is open in X. Let $U = f^{-1}(g^{-1}(O)$, hence U is neighbourhood of x. Therefore, $g \circ f$ is continuous. ■

Theorem 2.1.2. Suppose $f : X \to Y$ is continuous, let $A \subseteq X$ with topology τ, then (A, τ) is subspace topology. Then $f|_A : A \to Y$ is continuous.

Proof. Let O be an open set, $O \subseteq Y$, $(f|_A)^{-1}(O) = A \cap f^{-1}(O)$. Since f is continuous, $f^{-1}(O)$ is open in X. Therefore $(f|_A)^{-1}(O)$ is open in A. By claim, $f|_A$ is continuous. ■

Corollary 2.1.1. If $x \in A$, A is neighbourhood of x, then $f|_A$ is continuous and f is continuous on x.

Proof. Let V is neighbourhood of $f(X)$, exists $U_A \subset A$, it's open and such that $x \in U_A \subset f^{-1}|_A(V) = A \cap f^{-1}(V)$. Let $U_A = U \cap A$, U is open in X, then $U \cap \mathring{A}$ also open in X, implies $x \in U \cap \mathring{A} \subset U_A \subset f^{-1}(V)$. Therefore f is continuous. ■

Theorem 2.1.3. The following are equivalence:

(a). $f : X \to Y$ continuous.

(b). If $\mathcal{B}$ a basis of Y, then $f^{-1}(\mathcal{B})$ is open in X.

(c). $f(\overline{A}) \subseteq \overline{f(A)}$, $\forall A \subseteq X$.

(d). $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$, $\forall B \subseteq Y$.

(e). C is closed set in Y, $f^{-1}(C)$ closed in X.

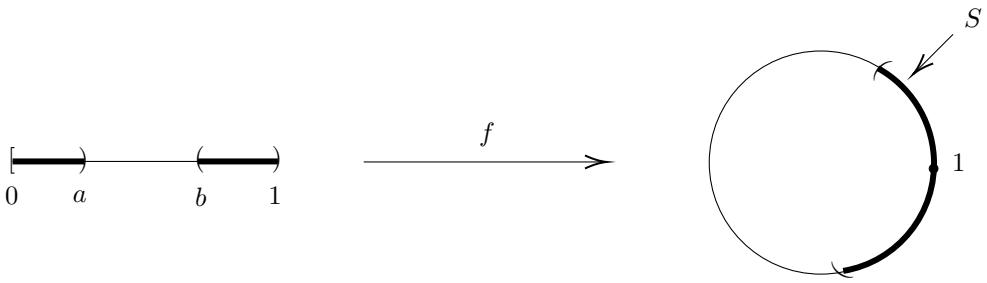
Proof. (a)$\Longrightarrow$(b): If $f : X \to Y$ continuous, by definition of basis, there's always exists $\mathscr{B}$ of $\mathcal{B}$ such that $x \in \mathscr{B} \subseteq \mathcal{B}$. Hence, we have $\bigcup_{i\in\mathcal{I}} \mathscr{B}_i = \mathcal{B}$ and $\mathscr{B}$ is open, so any finite union of open set is, then $\mathcal{B}$ is open in Y, by claim, $f^{-1}(\mathcal{B})$ is open in X.
(b)$\Longrightarrow$(c): Let $A \subseteq X$. Certainly, every point of $f(A)$ lies in $\overline{f(A)}$, therefore, we need show that if $x \in \overline{A} \setminus A$ and $f(x) \notin f(A)$, then point $f(x)$ is limit point of $f(A)$. If N is neighbourhood of $f(x)$ in in Y, we have basic open set $\mathscr{B} \subset \mathcal{B}$ such that $f(x) \in \mathcal{B} \subseteq N$, $f^{-1}(\mathcal{B})$ is open in X implies $f^{-1}(\mathscr{B}$ is open in X. Hence, $f^{-1}(\mathscr{B})$ is neighbourhood of x. But x is limit point of A. Therefore, $\mathscr{B}$ and N contains a point of $f(A)$, that's implies $f(\overline{A}) \subset \overline{f(A)}$.
(c)$\Longrightarrow$(d): If $f(\overline{A}) \subseteq \overline{f(A)}$, $\forall A \subseteq X$, for all $x \in \overline{f^{-1}(B)}$, by condition of (c), we have $f(x) \in f(\overline{f^{-1}(B)}) \subseteq \overline{ff^{-1}(B)} = \overline{B}$. Hence, $x \in f^{-1}(B)$ implies $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$.
(d)$\Longrightarrow$(e): If B is closed subset in Y, then $B = \overline{B}$, according (d), we have $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B}) = f^{-1}(B)$so $f^{-1}(B)$ is closed in X.
(e)$\Longrightarrow$(a): Let V be a neighbourhood of $f(x)$, $U = f^{-1}(V)$, by $f^{-1}(\mathring{V}) = (f^{-1}((\mathring{V})^c))^c$, because (e), $(\mathring{V})^c$ is closed, then $f^{-1}((\mathring{V})^c)$ is closed, so $(f^{-1}((\mathring{V})^c))^c$ is open, $x \in f^{-1}((\mathring{V}) \subset U$, hence, U is neighbourhood of x, therefore f is continuous. ■

Example. Let $\mathbb{S}^1$ be unit circle, taken subspace topology and give interval $[0, 1)$ induced topology from $\mathbb{R}$, define

$$\begin{aligned} f : [0, 1) &\longrightarrow \mathbb{S}^1 \\ x &\longmapsto e^{2\pi i t} \end{aligned}$$

It's clearly continuous. ◇

Example. Use above example, if we take the open segments of the circle as a basis for the topology on $\mathbb{S}$. Now let S be the segment, and if $1 \notin S$, where 1 is complex number, then $f^{-1}(S)$ is an open interval of the form (a, b) such that $a < a < b < 1$. So $f^{-1}(S)$ is open in $[0, 1)$. If S contain 1, then



$f^{-1}(S)$ has $[0, 1)\cup(b, 1)$ (it's intersection of $(-1, a)\cup(b, 1)$ of $\mathbb{R}$ with $[0, 1)$), by Theorem 2.1.3 (b), f is continuous and clearly bijection. ◇

Remark. f is not homeomorphism, we just need produced an open set O of $[0, 1)$ such that $(f^{-1})^{-1}(O) = f(O)$ is not open in $\mathbb{S}^1$. Take O to be the interval $[0, \frac{1}{2})$, this open in $[0, 1)$, its image $f([0, 1))$ under exp of those complex numbers $z \in \mathbb{S}^1$ for which $0 \leq \arg z < \pi$, it's not open.

Example. For X be any topological space, then there's a unique continuous function

- from the empty topological space to X

$$\emptyset \xrightarrow{\exists\,!} X$$

- from X to the point topological space

$$X \xrightarrow{\exists\,!} \{*\}$$

◇

Notation. The empty set there exists a unique topology τ making it a topological space. i.e. $\emptyset := (\emptyset, \tau_\emptyset = \{\emptyset\})$. We write $\{*\} := (\{1\}, \tau = \{\emptyset, 1\})$ be point topological space.

Example (Image factorization)**.** Let $f : X \to Y$ be a continuous function. Let $f(X \subset Y$ for the image of f on underlying sets, consider the resulting factorization of f through $f(X)$ on underlying sets, we have sequence:

$$f : X \twoheadrightarrow f(X) \hookrightarrow Y$$

To topologize the $\mathrm{Im}(f)$, consider the subspace topology from Y which evidently makes the inclusion $f(X \to Y$ a continuous function. Observe that this also makes $X \to f(X)$ a continuous function. An open subset of $f(X)$ in this case is of the form $U_Y \cap f(X)$ for $U_Y \in \tau_Y$, and $f^{-1}(U_Y \cap f(X)) = f^{-1}(U_Y)$ which is open in X since f continuous. ◇

Definition 2.1.2 (Embedding of topological spaces)**.** Let X and Y be topological spaces. A continuous function $f : X \to Y$ is called an embedding of topological spaces if in its image factorization, we have sequence

$$f : X \xrightarrow{\cong} f(X) \hookrightarrow Y$$

The image $f(X \hookrightarrow Y$ equipped with the subspace topology, we have $X \to f(X)$ is a homeomorphism.

Convention.

Proposition 2.1.1 ((open/closed continuous injections are embeddings)**.** A continuous function $f : X \longrightarrow Y$ satisfy:

- an injective function
- an open map or a closed map

Then it's an embedding of topological spaces.

Remark. This is called a closed embedding if the image $f(X \subset Y$ is a closed subset.

Proof of Proposition.

■

2.2 The Order Topology

2.3 Tietze Extension Theorem

Lecture 3

Compactness and Connectedness

3.1 Compact Spaces

3.2 Product Spaces

3.3 Connected Spaces

Lecture 4

Metric Space & Normed Linear Space

Lecture 5

Countability and Separation Axioms

Lecture 6

Identification Spaces

6.1 Relation and Equivalence relation

6.2 Quotient Topology

6.3 Topological Group

6.4 Orbit Space and Homogeneous Space